Approval-Based Committee Selection with Fairness Constraints

A Project Report Submitted in Partial Fulfillment of Requirements for the Degree of

Bachelor of Technology

by

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Abstract

Multiwinner voting is an important problem in computational social choice where the objective is to select a committee of candidates based on voter preferences. Traditional approval-based committee selection methods primarily focus on maximizing voter satisfaction and proportional representation but often fail to satisfy fairness and diversity requirements involving protected attributes such as gender, race, region, and category. In many real-world applications including hiring systems, recommendation systems, and scholarship allocation, committee selection must jointly optimize voter utility and fair representation.

In this project, we study approval-based committee selection under fairness constraints and analyze the tradeoff between utility maximization and fairness satisfaction. We investigate temporal committee selection frameworks, fairness-aware optimization models, and constrained committee construction under marginal quota requirements. We propose and analyze multiple frameworks including ILP-based optimization, Proportional Approval Fair Committee Selection (PAF-Comm), exact enumeration methods for binary attributes, and the Constraint-Integrated Greedy Approval Voting (CIGAV) algorithm for scalable fairness-aware committee selection. Experimental evaluation on synthetic and real-world datasets such as MovieLens and Goodreads demonstrates that the proposed methods significantly improve group representation while maintaining strong committee utility. The project highlights the importance of fairness-aware approval-based voting systems and provides scalable optimization techniques for constrained committee selection problems.

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Honor Code

We certify that we have properly cited any material taken from other sources and have obtained permission for any copyrighted material included in this report. We take full responsibility for any code submitted as part of this project and the contents of this report.

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Certificate

It is certified that the B. Tech. project "Approval-Based Committee Selection with Fairness Constraints" has been done by Shaikh Asra Swaleh(2022CSB1121), Radha (2022CSB1108) under supervision of Dr.Manish Kumar. This report has been submitted towards partial fulfillment of B. Tech. project requirements.

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Chapter 1

Introduction

1.1 Background and Motivation

Committee selection is a fundamental problem in computational social choice theory where the objective is to select a subset of candidates from a larger candidate pool based on voter preferences. Unlike single-winner voting systems, multiwinner voting focuses on selecting a committee of size k that best represents the preferences of the voters.

In many practical applications such as parliamentary elections, recommendation systems, hiring committees, scholarship selection, conference program committees, task allocation systems, and public decision-making processes, a group of candidates must be selected rather than a single winner. This motivates the study of multiwinner voting systems.

Traditional multiwinner voting rules include Approval Voting (AV), Proportional Approval Voting (PAV), Chamberlin–Courant Voting (CC), Satisfaction Approval Voting (SAV), and Greedy Approval Voting (GAV). These voting rules aim to maximize voter satisfaction, representation, or proportionality.

Approval Voting selects the k candidates with the highest number of approvals. Proportional Approval Voting improves proportional representation by using harmonic weighting of voter satisfaction. Chamberlin–Courant voting focuses on ensuring that every voter has a representative candidate in the committee. Satisfaction Approval Voting attempts to maximize overall voter satisfaction.

Although these classical voting rules perform well in unconstrained committee selection settings, they are insufficient for modern real-world committee construction problems. Practical committee selection problems frequently involve fairness requirements, diversity guarantees, and attribute-based representation constraints. For example, a hiring committee may require a minimum number of female candidates, candidates from different departments, or candidates belonging to underrepresented groups.

Similarly, recommendation systems may require diversity in genres, age groups, or regions. Scholarship committees may require balanced representation across categories and demographic groups. Traditional approval-based voting systems do not explicitly model these fairness-aware constraints.

Another important challenge is that modern committee selection problems often involve intersectional protected groups. A candidate may simultaneously belong to multiple protected categories such as gender, race, department, and region. Therefore, fairness constraints become highly interdependent and computationally difficult.

In addition, committee selection may occur repeatedly over time. In temporal committee selection settings, committees are selected over multiple rounds while preserving stability and fairness across stages. The selected committees should neither change too rapidly nor remain completely static.

These limitations motivate the study of fairness-aware approval-based committee selection frameworks that jointly optimize voter utility and group fairness under structural constraints.

1.2 Multiwinner Voting

Multiwinner voting refers to the process of selecting a committee of candidates from a larger candidate pool. Let

$$C = \{c_1, c_2, \ldots, c_m\}$$

be the set of candidates and

$$V = \{v_1, v_2, \ldots, v_n\}$$

be the set of voters.

Each voter v_i submits an approval ballot $A_i \subseteq C$ consisting of the candidates approved by the voter.

The objective is to select a committee

$$W \subseteq C$$

such that

$$|W| = k$$

where k denotes the committee size.

One of the simplest utility measures is the total approval utility:

$$U_{sum}(W) = \sum_{i=1}^{n} |A_i \cap W|$$

This utility simply counts the total number of approvals received by the selected committee.

However, simple approval utility may lead to domination by heavily represented voter groups. To address this issue, Proportional Approval Voting (PAV) introduces diminishing returns through harmonic weighting.

For a voter v_i, let

$$t_i(W) = |A_i \cap W|$$

Then the PAV utility of committee W is defined as

$$U_{PAV}(W) = \sum_{i=1}^{n} \sum_{j=1}^{t_i(W)} \frac{1}{j}$$

The harmonic weighting promotes proportional representation by reducing the marginal contribution of additional approved candidates for the same voter.

Multiwinner voting rules can therefore be viewed as optimization frameworks that attempt to maximize voter satisfaction under different representation objectives.

1.3 Fairness in Committee Selection

Fairness has become an increasingly important requirement in modern committee selection systems. In practical applications, candidates are often associated with protected attributes such as gender, race, age, department, region, or category.

Let

$$p_1(c), p_2(c), \ldots, p_q(c)$$

represent the protected attributes associated with candidate c.

For each attribute value, we define an attribute group:

$$G_{k,v} = \{c \in C : p_k(c) = v\}$$

Examples include:

- Male candidates
- Female candidates
- Candidates from a specific department
- Candidates from a specific region

In addition to attribute groups, fairness constraints may involve intersectional groups.

An intersectional group is formed by combining multiple attribute values simultaneously.

Examples include:

- Black Female
- White Male
- Senior Female Faculty

Intersectional groups are important because a candidate may simultaneously satisfy multiple fairness constraints.

The committee selection problem therefore becomes:

1. The committee satisfies fairness constraints.
2. The committee maximizes voter utility.
3. Representation is balanced across protected groups.

To model fairness, we define representation targets for each group.

For a group G, let r_G be the minimum required representation.

The representation shortfall is defined as

$$short(G, W) = \max(0, r_G - |W \cap G|)$$

If the selected committee satisfies the group requirement, the shortfall becomes zero.

We also define group satisfaction as

$$U_G(W) = \sum |A_i \cap W \cap G|$$

which measures how well the committee represents a specific group.

The fairness objective may therefore involve:

- minimizing representation shortfall,
- maximizing group satisfaction,
- maximizing minimum group satisfaction.

These fairness-aware objectives introduce significant computational complexity into committee selection.

1.4 Motivation for Fairness Constraints

Traditional multiwinner voting rules primarily optimize voter satisfaction and proportionality. However, these methods fail to explicitly model realistic fairness and diversity requirements.

In many real-world scenarios, fairness constraints are externally imposed by organizations, governments, or institutional policies. Examples include:

- Minimum female representation in hiring panels
- Regional diversity in scholarship selection
- Departmental diversity in academic committees
- Fair genre representation in recommendation systems

A major challenge arises because the committee maximizing approval utility may violate fairness constraints.

For example, suppose a highly approved group of candidates all belong to the same demographic category. Selecting only these candidates may maximize approval utility but produce unfair representation.

Conversely, satisfying strict fairness constraints may require selecting less popular candidates, thereby reducing voter satisfaction.

This creates an approval–constraint tension between:

- maximizing approval utility,
- satisfying fairness constraints.

Furthermore, fairness constraints become increasingly difficult when protected attributes overlap. A candidate may contribute simultaneously to multiple quotas, making the optimization problem highly interdependent.

As the number of protected attributes increases, the committee selection problem becomes computationally difficult and often NP-hard.

These challenges motivate the development of fairness-aware optimization frameworks that jointly balance utility maximization and fairness satisfaction.

1.5 Contributions of the Project

The major contributions of this project are summarized below:

- We studied classical and temporal multiwinner committee selection frameworks.
- We analyzed welfare–fairness tradeoffs in approval-based committee construction.
- We formulated Integer Linear Programming (ILP)-based optimization models for fairness-aware committee selection.
- We proposed a Proportional Approval Fair Committee Selection framework called PAF-Comm.
- We introduced fairness objectives based on representation shortfall and max-min group satisfaction.
- We studied committee selection under marginal attribute quota constraints.
- We developed an exact enumeration algorithm for committee selection with binary protected attributes.
- We analyzed the computational complexity and NP-hardness of generalized fairness-aware committee selection.
- We proposed a Constraint-Integrated Greedy Approval Voting (CIGAV) algorithm for scalable fairness-aware committee selection.
- We experimentally evaluated the proposed frameworks on synthetic and real-world datasets including MovieLens and Goodreads datasets.
- We compared fairness-aware committee selection methods with classical voting rules using fairness and welfare metrics.

1.6 Organization of the Report

The remainder of this report is organized as follows.

Chapter 2 presents the literature review and discusses existing work related to multiwinner voting, temporal committee selection, fairness-aware voting, and constrained committee construction.

Chapter 3 formally defines the fairness-aware approval-based committee selection problem and introduces mathematical formulations for utility and fairness objectives.

Chapter 4 presents the proposed algorithms and optimization frameworks including ILP-based optimization, PAF-Comm, exact enumeration methods, and the CIGAV algorithm.

Chapter 5 presents the experimental setup, datasets, evaluation metrics, and experimental results.

Finally, Chapter 6 concludes the report and discusses limitations and future research directions.

Chapter 2

Literature Review

2.1 Classical Multiwinner Voting Rules

Multiwinner voting has been extensively studied in computational social choice theory. Traditional multiwinner voting systems aim to select a committee that best represents voter preferences.

Approval Voting (AV) is one of the simplest approval-based voting systems. In AV, each voter approves any number of candidates, and the k candidates with the highest approval scores are selected. Although AV is computationally simple and easy to implement, it does not provide proportional representation.

Proportional Approval Voting (PAV) was introduced to improve proportional representation in committee selection. PAV uses harmonic weighting to reduce the dominance of heavily represented voter groups. By applying diminishing returns to repeated representation, PAV promotes balanced committee construction.

Chamberlin–Courant Voting (CC) focuses on representative coverage. The goal is to ensure that every voter has at least one representative candidate in the committee. CC-based rules are effective for representation but are computationally expensive.

Satisfaction Approval Voting (SAV) attempts to maximize total voter satisfaction by considering the fraction of approved candidates selected for each voter.

Greedy Approval Voting (GAV) is a scalable approximation-based approach that greedily constructs committees while attempting to preserve representation.

Although these classical approaches provide effective committee selection mechanisms, they do not explicitly model fairness constraints related to candidate attributes.

2.2 Temporal Committee Selection

In many practical applications, committees are selected repeatedly over time rather than in a single round. Examples include weekly scheduling, rotating panels, recurring recommendation systems, and sequential task allocation.

Temporal committee selection introduces additional challenges:

- stability across consecutive committees,
- controlled adaptation over time,
- fairness across multiple rounds.

Conservative temporal models aim to preserve committee stability by minimizing changes between consecutive committees.

Revolutionary temporal models enforce significant changes between stages to encourage diversity and rotation.

Temporal committee selection frameworks therefore attempt to balance:

- committee quality,
- stability,
- diversity,
- fairness.

Successive committee election models additionally impose frequency constraints that restrict how many times a candidate may appear across committee sequences.

These temporal frameworks demonstrate that fairness-aware committee selection is not limited to a single election but extends naturally to multi-stage decision-making environments.

2.3 Fair Committee Selection

Fairness-aware committee selection has recently emerged as an important research area due to growing concerns regarding representation and diversity.

Existing research introduces fairness constraints in the form of:

- group quotas,
- diversity constraints,
- representation guarantees,
- parity conditions.

Many fairness-aware voting frameworks attempt to ensure that protected groups are adequately represented in the final committee.

Group fairness constraints typically require that a minimum number of selected candidates belong to specific demographic groups.

Intersectional fairness further extends this framework by considering combinations of protected attributes simultaneously.

Recent work also studies fairness in terms of group satisfaction rather than only numerical representation.

In these frameworks, fairness is evaluated based on how well each group is served by the selected committee.

However, several existing fairness-aware approaches rely heavily on ranking-based systems or pairwise comparisons.

In contrast, approval-based fairness frameworks operate directly on approval ballots and committee-level optimization.

This motivates the study of approval-native fairness-aware committee selection frameworks.

2.4 Constraint-Based Committee Selection

Constraint-based committee selection focuses on selecting committees that satisfy externally imposed structural constraints.

Marginal quota constraints are among the most widely studied fairness constraints.

Under marginal quotas, each protected attribute value is associated with a minimum representation requirement.

For example:

- at least 3 female candidates,
- at least 2 candidates from department A,
- at least 4 senior members.

When candidates possess multiple binary attributes, every candidate belongs to a specific type.

For two binary attributes, candidates can be partitioned into four types:

- C_{00}
- C_{01}
- C_{10}
- C_{11}

A committee is feasible if it satisfies all quota requirements simultaneously.

Exact enumeration methods can be used to explore feasible count vectors and construct optimal committees.

Although binary-attribute committee selection can sometimes be solved in polynomial time, the generalized multi-attribute problem becomes NP-hard.

Constraint-based committee selection therefore introduces significant computational and combinatorial challenges.

2.5 Research Gap

Although significant progress has been made in multiwinner voting and fairness-aware committee selection, several limitations remain.

First, many classical voting systems optimize voter satisfaction without considering fairness constraints.

Second, fairness-aware methods often focus only on representation counts and ignore approval utility.

Third, several existing approaches rely on ranking-based systems rather than approval ballots.

Fourth, intersectional fairness constraints significantly increase computational complexity and are difficult to handle efficiently.

Fifth, exact optimization approaches such as ILP become computationally expensive for large-scale datasets.

Finally, there is limited work on scalable fairness-aware committee selection algorithms that jointly optimize utility and fairness.

The work presented in this project attempts to address these limitations through approval-based fairness modeling, exact optimization, scalable greedy algorithms, and fairness-aware utility formulations.

Chapter 3

Problem Formulation

3.1 Basic Definitions

Let

$$C = \{c_1, c_2, \ldots, c_m\}$$

be the set of candidates and

$$V = \{v_1, v_2, \ldots, v_n\}$$

be the set of voters.

Each voter v_i submits an approval ballot

$$A_i \subseteq C$$

where $c \in A_i$ indicates that voter v_i approves candidate c.

A committee is a subset

$$W \subseteq C$$

such that

$$|W| = k$$

where k is the committee size.

The objective is to construct a committee that maximizes voter utility while satisfying fairness constraints.

3.2 Approval Utility

The approval score of a candidate c is defined as

$$score(c) = \sum 1[c \in A_i]$$

which counts the number of voters approving candidate c.

The total approval utility of a committee W is

$$U_{sum}(W) = \sum score(c)$$

where the summation is taken over all candidates in W.

To improve proportional representation, we use Proportional Approval Voting utility.

For a voter v_i,

$$t_i(W) = |A_i \cap W|$$

The PAV utility is defined as

$$U_{PAV}(W) = \sum_{i=1}^{n} \sum_{j=1}^{t_i(W)} \frac{1}{j}$$

The harmonic weighting ensures diminishing returns for additional approved candidates belonging to the same voter.

3.3 Protected Attributes and Groups

Each candidate is associated with multiple protected attributes.

For an attribute p_k and value v, the corresponding attribute group is

$$G_{k,v} = \{c \in C : p_k(c) = v\}$$

Examples include:

- Female candidates
- Candidates from region A
- Senior faculty members

Intersectional groups are defined by combining multiple attribute values simultaneously.

Examples include:

- Black Female
- White Male
- Senior Female Faculty

A candidate may belong to multiple protected groups simultaneously.

This overlap creates interdependent fairness constraints.

3.4 Representation Constraints

For each protected group G, let

$$r_G$$

represent the minimum required representation.

The representation shortfall of group G is defined as

$$short(G, W) = \max(0, r_G - |W \cap G|)$$

If the committee satisfies the representation requirement, the shortfall becomes zero.

The total representation penalty is defined as

$$Pen_{rep}(W) = \sum short(G, W)$$

We also define group satisfaction as

$$U_G(W) = \sum |A_i \cap W \cap G|$$

The normalized group satisfaction is

$$\tilde{U}_G(W) = \frac{U_G(W)}{U_G^{max}}$$

where U_G^{max} denotes the maximum achievable group utility.

The max-min group satisfaction objective is defined as

$$MM(W) = \min \tilde{U}_G(W)$$

This objective attempts to maximize the welfare of the least satisfied group.

3.5 Temporal Committee Selection Formulation

In temporal committee selection, a sequence of committees is selected over multiple stages.

Let

$$S = (S_1, S_2, \dots, S_t)$$

represent a committee sequence.

Each committee satisfies

$$|S_i| = k$$

for all stages i.

Temporal committee selection introduces stability constraints between consecutive committees.

Conservative models attempt to minimize committee changes:

$$|S_i \cap S_{i+1}| \geq x$$

Revolutionary models encourage significant changes:

$$|S_i \cap S_{i+1}| \leq y$$

Temporal frameworks therefore balance:

- utility,
- diversity,
- stability,
- fairness.

3.6 Marginal Quota Constraints

Suppose each candidate possesses two binary attributes:

$$X(c), Y(c) \in \{0, 1\}$$

Candidates are partitioned into four types:

- C_{00}
- C_{01}
- C_{10}
- C_{11}

A committee W is feasible if

$$|W| = k$$

and all quota constraints are satisfied.
For example:

$$|W \cap C_{X=0}| \geq q_{X=0}$$

$$|W \cap C_{Y=1}| \geq q_{Y=1}$$

A candidate may contribute simultaneously to multiple quotas.

The committee can also be represented using a type-count vector:

$$x = (x_{00}, x_{01}, x_{10}, x_{11})$$

where x_{ab} denotes the number of selected candidates from type C_{ab}.

3.7 Approval–Constraint Tension

A major challenge in fairness-aware committee selection is the conflict between approval utility and fairness satisfaction.

The committee maximizing approval utility may violate fairness constraints.

Conversely, satisfying all fairness constraints may require selecting candidates with lower approval support.

Therefore, fairness-aware committee selection naturally involves a tradeoff between:

- utility maximization,
- fairness satisfaction.

This approval–constraint tension forms the central challenge of constrained committee selection.

3.8 Complexity Analysis

Several fairness-aware committee selection problems are computationally difficult.

Maximizing PAV utility is NP-hard.

Fair committee selection under multiple intersectional constraints is also NP-hard.

When the number of protected attributes becomes part of the input, the generalized marginal quota problem becomes NP-hard through reductions from Set Cover.

Exact optimization therefore becomes computationally expensive for large-scale committee selection settings.

This motivates the study of scalable approximation and greedy optimization algorithms.

Chapter 4

Proposed Methods and Algorithms

4.1 ILP-Based Optimization Framework

To jointly optimize voter utility and fairness, we formulate committee selection as an Integer Linear Programming (ILP) problem.

Let

$$x_i = \begin{cases} 1, & \text{if candidate } c_i \text{ is selected} \\ 0, & \text{otherwise} \end{cases}$$

The committee size constraint is defined as

$$\sum x_i = k$$

where k denotes the committee size.

The fairness constraints are represented as

$$\sum_{c_i \in G} x_i \geq r_G$$

for every protected group G, where r_G denotes the minimum required representation for group G.

The optimization objective is

$$\text{Maximize } \; Utility - \lambda(FairnessPenalty)$$

where λ controls the tradeoff between utility maximization and fairness satisfaction.

The ILP framework enables exact optimization of committee selection under fairness constraints. However, ILP-based optimization becomes computationally expensive for large-scale datasets due to the combinatorial nature of the optimization problem.

4.1.1 Utility Maximization

The utility component of the optimization framework measures voter satisfaction.

Approval utility is computed using approval ballots submitted by voters. The optimization attempts to maximize the total approval utility of the selected committee while satisfying structural fairness constraints.

To improve proportional representation, harmonic weighting functions such as Proportional Approval Voting utility are incorporated into the optimization objective.

4.1.2 Fairness Constraints

Fairness constraints ensure adequate representation for protected groups.

These constraints may involve:

- gender representation,
- regional diversity,
- departmental diversity,
- intersectional fairness.

The optimization framework jointly balances utility and fairness through penalty-based formulations.

4.1.2.1 Representation Penalties

Representation penalties are introduced whenever the committee violates group representation requirements.

The penalty-based formulation allows soft enforcement of fairness constraints while preserving flexibility in optimization.

4.2 PAF-Comm Framework

To improve proportional fairness in approval-based committee selection, we propose the Proportional Approval Fair Committee Selection (PAF-Comm) framework.

The framework combines:

- PAV utility,
- representation shortfall penalties,
- max-min group satisfaction.

The optimization objective is defined as

$$F_{PAF}(W) = U(W) - \lambda Pen_{rep}(W)$$

where:

- $U(W)$ denotes PAV utility,
- $Pen_{rep}(W)$ denotes representation penalty,
- λ controls fairness importance.

We also incorporate max-min group satisfaction:

$$MM(W) = \min \tilde{U}_G(W)$$

The committee is constructed greedily.

At every step, the algorithm selects the candidate providing the best combined improvement in:

- utility,
- fairness,
- group satisfaction.

The greedy framework provides scalable fairness-aware committee construction while maintaining high approval utility.

4.2.1 Representation Shortfall

The representation shortfall measures the extent to which a protected group is underrepresented in the selected committee.

For a protected group G, the shortfall is defined as

$$short(G, W) = \max(0, r_G - |W \cap G|)$$

Lower shortfall values indicate improved fairness satisfaction.

4.2.2 Max-Min Group Satisfaction

The max-min objective attempts to maximize the welfare of the least satisfied group.

This prevents situations where one protected group receives disproportionately low representation or satisfaction compared to other groups.

4.2.2.1 Greedy Committee Construction

The PAF-Comm framework constructs committees iteratively.

At every iteration:

1. Candidate utility contribution is evaluated.
2. Fairness improvement is computed.
3. Group satisfaction impact is analyzed.
4. The best candidate is selected greedily.

This process continues until the committee size reaches k.

4.3 Exact Enumeration Algorithm for Marginal Quotas

For committee selection with two binary attributes, candidates are partitioned into four types:

- C_{00}
- C_{01}
- C_{10}
- C_{11}

A committee is represented using a count vector:

$$x = (x_{00}, x_{01}, x_{10}, x_{11})$$

where x_{ab} denotes the number of selected candidates from group C_{ab}.

The algorithm enumerates all feasible count vectors satisfying:

- committee size constraints,
- quota constraints.

For every feasible count vector, the top-scoring candidates from each type are selected.

The committee with maximum approval utility is returned.

The algorithm proceeds as follows:

1. Compute approval scores.
2. Partition candidates into attribute groups.
3. Sort each group by approval score.
4. Enumerate feasible count vectors.
5. Construct committees for feasible vectors.

6. Return the best feasible committee.

The total running time is polynomial for the binary-attribute setting. However, the generalized multi-attribute problem becomes NP-hard.

4.3.1 Feasible Count Vectors

A count vector is feasible if:

- the total committee size equals k,
- all quota constraints are satisfied.

Feasible count vectors significantly reduce the committee search space.

4.3.2 Committee Construction

Once a feasible count vector is identified, candidates are selected greedily according to approval scores within each candidate type.

This guarantees maximum approval utility for the corresponding feasible configuration.

4.3.2.1 Complexity Analysis

The enumeration algorithm remains polynomial-time for two binary attributes.

However, the number of candidate types grows exponentially with the number of protected attributes, leading to NP-hardness in the generalized setting.

4.4 Constraint-Integrated Greedy Approval Voting (CIGAV)

To improve scalability, we study a greedy fairness-aware committee construction algorithm called Constraint-Integrated Greedy Approval Voting (CIGAV).

CIGAV dynamically balances:

- approval utility,

- fairness urgency.

The algorithm uses two important components:

1. Feasibility checking
2. Constraint urgency modeling

A partial committee is feasible if it can still be extended into a complete committee satisfying all fairness constraints.

Constraint urgency measures how critical a fairness constraint is relative to remaining committee slots.

The urgency of a constraint is defined as

$$Urgency = \frac{RemainingQuota}{RemainingSlots}$$

The overall pressure is the maximum urgency among all constraints.

The candidate scoring function combines:

- approval score,
- fairness urgency contribution.

The score of a candidate is defined as

$$Score(c_i) = \alpha Approval(c_i) + \beta ConstraintContribution(c_i)$$

where:

- $\alpha = 1 - \beta$
- β represents fairness pressure.

As the algorithm approaches the final committee slots, fairness pressure increases automatically.

Thus, the algorithm dynamically shifts from approval optimization toward fairness satisfaction.

The algorithm proceeds as follows:

1. Compute approval scores.
2. Compute constraint satisfaction vectors.
3. Initialize an empty committee.
4. Compute fairness pressure.
5. Evaluate candidate scores.
6. Perform feasibility-safe greedy selection.
7. Update quotas.
8. Repeat until committee completion.

CIGAV provides scalable committee selection while maintaining fairness guarantees.

4.4.1 Feasibility Checking

Feasibility checking ensures that partial committee selections can still be extended into feasible complete committees.

If adding a candidate makes future fairness satisfaction impossible, the candidate is rejected during greedy selection.

4.4.2 Constraint Urgency Modeling

Constraint urgency determines the importance of satisfying a fairness constraint relative to remaining committee slots.

Higher urgency values force the algorithm to prioritize fairness-aware candidate selection.

4.4.2.1 Adaptive Greedy Optimization

The adaptive weighting mechanism dynamically balances utility and fairness during committee construction.

Initially, approval utility dominates candidate selection. As fairness constraints become tighter, fairness pressure increases automatically.

4.5 Local Search Refinement

To further improve committee quality, local search refinement strategies are applied.

The local search process attempts candidate substitutions and committee swaps.

The objective is to estimate the price of fairness and improve committee utility while preserving fairness constraints.

At every iteration:

- one candidate is removed,
- another candidate is inserted,
- committee quality is re-evaluated.

Local search refinement improves fairness–utility tradeoffs and reduces suboptimal committee construction.

4.6 Complexity and Approximation Guarantees

The exact enumeration algorithm is polynomial-time for binary attributes.

However, generalized fairness-aware committee selection becomes NP-hard.

The CIGAV algorithm provides scalable approximation-based optimization.

The approximation ratio depends on:

- total quota demand,
- committee size.

When quota demand is small relative to committee size, the approximation quality improves.

Greedy optimization significantly reduces computational overhead compared to exact optimization methods.

Approximation-based methods therefore provide practical scalability for large constrained committee selection settings.

Chapter 5

Experimental Analysis and Results

5.1 Experimental Setup

The proposed fairness-aware committee selection frameworks were evaluated on both synthetic and real-world datasets.

The experiments were designed to analyze:

- approval utility,
- fairness satisfaction,
- runtime efficiency,
- scalability.

Three categories of datasets were used:

- Synthetic datasets
- MovieLens dataset
- Goodreads Books dataset

The synthetic dataset contains:

- 6 candidates,
- 7 voters,
- committee size $k = 4$.

The protected groups include:

- Male/Female,
- White/Black,
- Black Male,
- White Female,
- Black Female.

The MovieLens dataset contains:

- 1682 movies,
- 943 approval ballots.

Protected attributes include:

- genre,
- year group,
- intersectional groups.

The Goodreads dataset contains:

- 10,000 books,
- 51,296 approval ballots.

Protected attributes include:

- year group,
- popularity group,

- author group.

The experiments compare the following committee selection methods:

- Approval Voting (AV),
- Proportional Approval Voting (PAV),
- Chamberlin–Courant (CC),
- ILP Optimization,
- PAF-Comm,
- CIGAV.

5.1.1 Synthetic Dataset Configuration

The synthetic dataset was specifically designed to evaluate fairness-aware committee selection under controlled conditions.

The dataset allows systematic analysis of:

- fairness penalties,
- utility degradation,
- quota satisfaction,
- intersectional constraints.

Different fairness parameter settings were evaluated to observe the tradeoff between voter utility and fairness satisfaction.

5.1.2 Real-World Datasets

The MovieLens and Goodreads datasets were selected because they naturally contain:

- large candidate pools,

- user preference information,
- diverse attribute groups,
- recommendation-based approval behavior.

These datasets help evaluate the scalability and practical applicability of fairness-aware committee selection frameworks.

5.2 Evaluation Metrics

Several evaluation metrics were used to analyze committee quality and fairness performance.

5.2.1 Utility Metrics

The following utility metrics were evaluated:

- approval utility,
- PAV utility,
- representation quality.

Approval utility measures the total number of voter approvals received by the selected committee.

PAV utility measures proportional representation using harmonic weighting.

Representation quality evaluates how effectively the committee represents voter preferences.

5.2.2 Fairness Metrics

The following fairness metrics were evaluated:

- representation shortfall,
- normalized group satisfaction,

- max-min fairness.

Representation shortfall measures the extent to which protected group quotas remain unsatisfied.

Normalized group satisfaction evaluates fairness across different protected groups.

Max-min fairness measures the welfare of the least satisfied protected group.

5.2.2.1 Efficiency Metrics

The following efficiency metrics were analyzed:

- runtime,
- scalability,
- approximation quality.

These metrics evaluate the computational efficiency and scalability of fairness-aware optimization frameworks.

5.3 Synthetic Dataset Results

The synthetic dataset experiments demonstrate the effect of fairness parameters on committee construction.

The λ parameter controls the importance of representation penalties.

As λ increases:

- fairness improves,
- utility slightly decreases.

Similarly, the μ parameter controls the importance of max-min group satisfaction.

The experimental graphs show that:

- stronger fairness constraints improve group representation,

- utility degradation remains moderate,
- fairness-aware optimization produces balanced committees.

The experiments also demonstrate that fairness-aware methods outperform classical voting rules under constrained settings.

5.3.1 Effect of Fairness Penalties

Higher fairness penalties force the optimization framework to prioritize underrepresented groups.

Although utility decreases slightly, the resulting committees become significantly more balanced and representative.

5.3.2 Tradeoff Between Utility and Fairness

The experiments clearly demonstrate the existence of an approval–constraint tradeoff.

Committees maximizing approval utility often fail to satisfy fairness constraints completely.

Conversely, committees satisfying stronger fairness requirements may experience slight reductions in approval utility.

5.3.2.1 Balanced Committee Construction

The proposed frameworks successfully balance:

- voter satisfaction,
- fairness,
- proportional representation.

The resulting committees maintain high utility while significantly improving fairness satisfaction.

5.4 Real Dataset Results

Experiments on MovieLens and Goodreads datasets demonstrate the scalability of the proposed methods.

The MovieLens dataset experiments show that fairness-aware committee selection improves genre diversity and temporal representation.

The Goodreads dataset experiments demonstrate that the proposed algorithms can handle large-scale committee selection under multi-level and intersectional constraints.

The experiments further show that:

- greedy optimization scales effectively,
- fairness constraints remain feasible,
- representation quality improves significantly.

5.4.1 MovieLens Dataset Analysis

The MovieLens experiments demonstrate that fairness-aware recommendation committees improve:

- genre diversity,
- representation balance,
- recommendation fairness.

The experiments also show that fairness-aware optimization reduces overrepresentation of dominant movie categories.

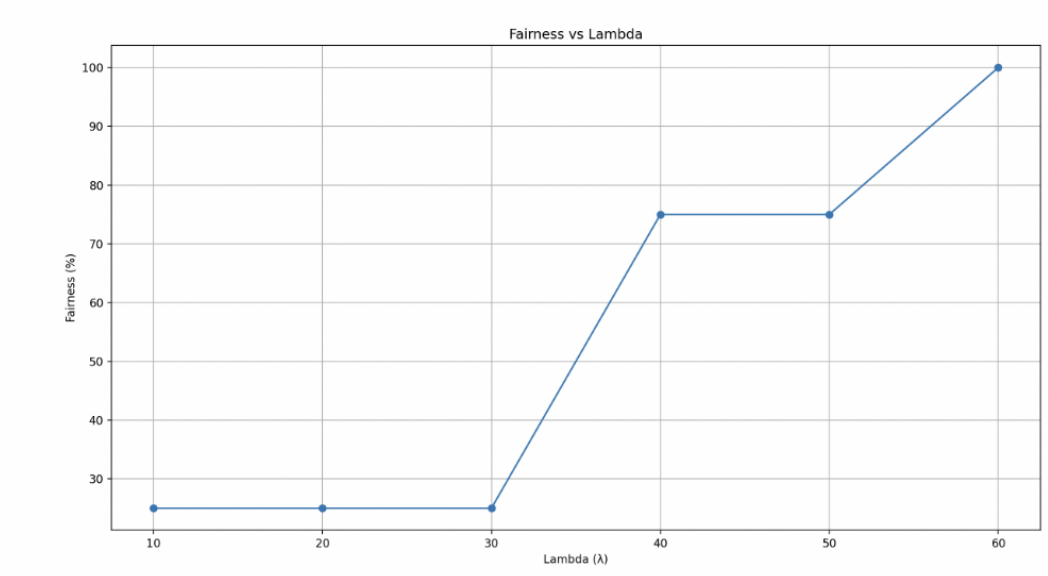

Figure 5.1: Effect of fairness penalty parameter λ on fairness satisfaction in MovieLens dataset experiments.

5.4.2 Goodreads Dataset Analysis

The Goodreads dataset experiments evaluate committee selection at significantly larger scale.

The proposed algorithms successfully handle:

- large candidate pools,
- multiple attribute groups,
- intersectional fairness constraints.

Scalable greedy optimization methods such as CIGAV achieve strong fairness performance while maintaining computational efficiency.

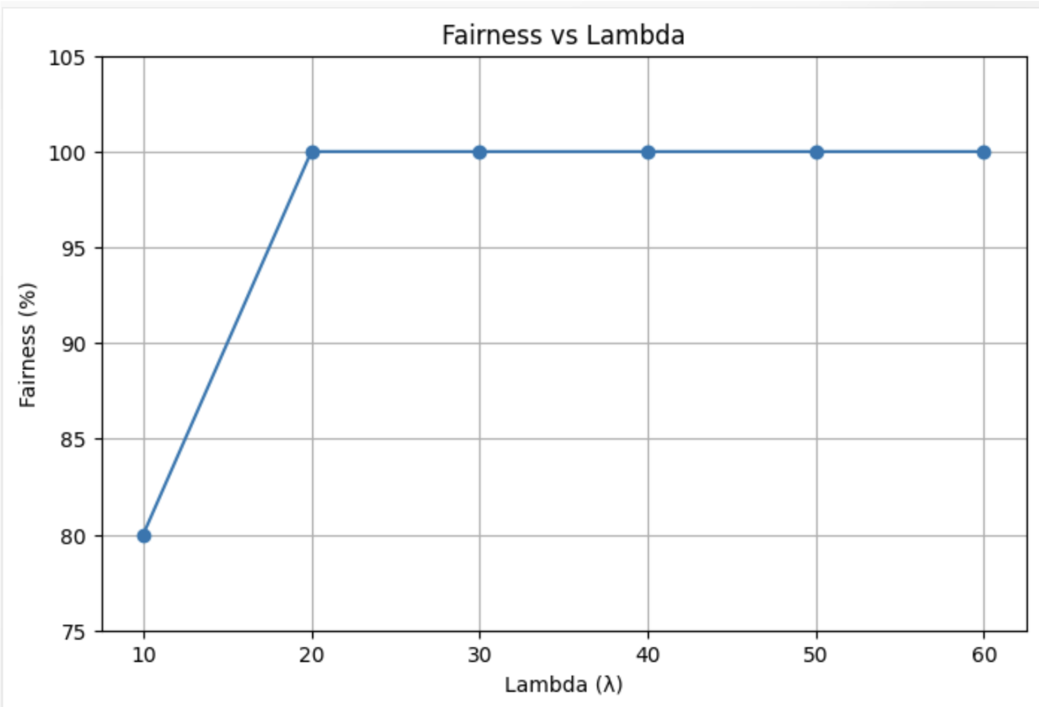

Figure 5.2: Effect of fairness penalty parameter λ on fairness satisfaction in Goodreads dataset experiments.

5.4.2.1 Scalability Evaluation

The experiments show that greedy fairness-aware optimization methods scale significantly better than exact ILP-based optimization methods for large datasets.

Approximation-based optimization therefore provides practical applicability for real-world fairness-aware committee selection systems.

5.5 Comparative Analysis

Approval Voting achieves high utility but poor fairness.

PAV improves proportionality but does not explicitly enforce fairness constraints.

ILP optimization provides strong fairness guarantees but suffers from scalability limitations.

PAF-Comm balances approval utility and fairness effectively.

CIGAV provides scalable fairness-aware committee construction while maintaining strong committee quality.

The comparative analysis demonstrates that fairness-aware optimization frameworks outperform classical methods in constrained committee selection settings.

5.5.1 Comparison of Classical and Fairness-Aware Methods

Classical voting rules primarily focus on utility maximization and proportional representation.

However, they frequently violate structural fairness requirements under constrained settings.

Fairness-aware optimization methods explicitly incorporate fairness constraints into committee construction and therefore produce more balanced committees.

5.5.2 Performance of PAF-Comm and CIGAV

PAF-Comm achieves strong fairness guarantees while preserving high approval utility.

CIGAV significantly improves scalability through adaptive greedy optimization and feasibility-aware selection.

The experiments demonstrate that CIGAV achieves near-optimal fairness performance while reducing computational complexity.

5.5.2.1 Runtime Comparison

Exact ILP-based optimization becomes computationally expensive for large datasets.

Greedy approximation-based methods scale more efficiently and remain practical for large constrained committee selection problems.

5.6 Observations

The experimental analysis leads to several important observations.

1. Increasing fairness requirements slightly reduces approval utility.
2. Greedy fairness-aware optimization scales significantly better than exact optimization.
3. Intersectional constraints increase computational complexity.
4. Dynamic weighting improves committee feasibility.
5. Fairness-aware committee selection significantly improves protected-group representation.
6. Utility loss due to fairness constraints remains manageable in most practical settings.

Overall, the experiments demonstrate that fairness-aware committee selection frameworks effectively balance voter satisfaction and representation fairness while maintaining scalability for real-world applications.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This project studied approval-based committee selection under fairness constraints.

We analyzed classical multiwinner voting frameworks and identified their limitations in fairness-sensitive applications.

To address these limitations, we studied fairness-aware optimization frameworks including ILP-based optimization, PAF-Comm, exact enumeration methods, and the Constraint-Integrated Greedy Approval Voting (CIGAV) algorithm.

The proposed frameworks jointly optimize:

- approval utility,
- proportional representation,
- fairness satisfaction.

We also analyzed temporal committee selection, intersectional fairness constraints, and approval–constraint tradeoffs.

Experimental evaluation on synthetic and real-world datasets demonstrated that fairness-aware committee selection can significantly improve protected-group representation while maintaining strong committee quality.

The results show that scalable fairness-aware optimization is achievable even under complex quota constraints.

Overall, the project highlights the importance of integrating fairness into approval-based committee selection systems and provides both exact and scalable approaches for practical constrained committee construction.

6.1.1 Summary of Contributions

The major contributions of the project are summarized below:

- Study of temporal and fairness-aware multiwinner voting frameworks.
- Development of ILP-based optimization formulations.
- Analysis of approval–constraint tradeoffs.
- Proposal of the PAF-Comm framework for fairness-aware approval voting.
- Development of exact enumeration methods for marginal quota constraints.
- Proposal of the scalable CIGAV framework for constrained committee construction.
- Experimental evaluation on synthetic and real-world datasets.

6.1.2 Impact of the Work

The proposed frameworks demonstrate that fairness-aware committee selection can be integrated effectively into real-world systems such as:

- recommendation systems,
- hiring systems,
- scholarship allocation,
- panel selection,
- public decision-making systems.

The work also highlights the importance of balancing utility maximization with fairness guarantees in computational social choice systems.

6.2 Limitations

Although the proposed frameworks perform effectively, several limitations remain.

1. Exact optimization methods become computationally expensive for large-scale datasets.
2. Generalized multi-attribute committee selection remains NP-hard.
3. Intersectional fairness constraints significantly increase optimization complexity.
4. Temporal committee selection introduces additional scalability challenges.
5. Real-world datasets may contain noisy or incomplete approval preferences.
6. Approximation-based methods may not always achieve globally optimal committees.

These limitations indicate that fairness-aware committee selection remains a computationally challenging optimization problem.

6.2.1 Scalability Challenges

As the number of candidates, voters, and protected attributes increases, exact optimization methods become increasingly expensive.

Large-scale real-world systems therefore require scalable approximation and greedy optimization techniques.

6.2.2 Fairness Complexity

Intersectional fairness introduces highly interdependent quota constraints.

Satisfying multiple overlapping fairness requirements simultaneously remains one of the major challenges in constrained committee selection.

6.3 Future Work

Several important directions remain for future research.

1. Develop stronger feasibility-preserving optimization methods for intersectional fairness constraints.

2. Study parameterized and approximation algorithms with tighter theoretical guarantees.

3. Extend fairness-aware committee selection to dynamic and online environments.

4. Investigate probabilistic fairness and uncertainty-aware committee selection.

5. Explore machine learning assisted committee scoring and adaptive fairness modeling.

6. Study streaming committee selection under continuously changing voter preferences.

7. Develop scalable distributed optimization methods for large-scale constrained committee selection.

8. Investigate fairness-aware recommendation systems and real-time decision-making applications.

6.3.1 Dynamic and Online Committee Selection

Future work may extend fairness-aware committee construction to dynamic environments where:

- voters arrive continuously,
- candidate attributes evolve over time,
- committee requirements change dynamically.

Such settings require adaptive and online optimization algorithms.

6.3.2 Learning-Based Fairness Optimization

Machine learning techniques may be incorporated into fairness-aware committee selection systems to:

- predict voter preferences,
- estimate fairness tradeoffs,
- adaptively tune optimization parameters.

Learning-assisted optimization can significantly improve scalability and personalization in practical recommendation systems.

6.3.3 Scalable Distributed Optimization

Future large-scale committee selection systems may require distributed and parallel optimization techniques.

Distributed fairness-aware optimization can improve:

- scalability,
- runtime efficiency,
- real-time committee construction.

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